

V4.2 Toy Model 5B

Graph Edge Economics Derived Relational Tension

Rev A - Dark Sector / Rotation-Curve Derivation Test

Date: 2026.05.26 | Continuation after Toy Model 5A

Status: Toy Model 5B replaces the assumed 5A flat-speed ansatz with a derivation from a radial graph action. It shows that an effectively 2D annular relational graph naturally produces an outer $1/r$ acceleration and therefore a flat rotation contribution. It does not derive the coupling amplitude, fit real galaxies, solve lensing/CMB/structure formation, disprove particle dark matter, or prove physical 3T.

1. Purpose

Toy Model 5A assumed a relational-tension contribution that flattened the rotation curve. Toy Model 5B removes that assumption. It starts from graph edge economics: a radial annular network tries to minimize uneven edge response while coupling to a central visible source.

The narrow question is: does the equilibrium of that network produce an outer acceleration proportional to $1/r$, which is the acceleration law needed for a flat rotation curve?

2. Native radial graph action

The graph is treated as an effective 2D annular disk. The response field $\Phi(r)$ is governed by the radial action:

$$A[\Phi] = 2 \pi \int r \left[\frac{1}{2} \left(\frac{d\Phi}{dr} \right)^2 - S(r) \Phi(r) \right] dr$$

Meaning of symbols:

- $\Phi(r)$ is the radial relational response field.
- $S(r)$ is the visible/baryonic relational source distribution.
- r is the annular radius.
- The factor $2 \pi r$ is the circumference weight: outer rings contain more relational edge channels.
- The gradient term penalizes uneven response between neighboring radial rings.
- The source term rewards response where the visible source exists.

3. Variation and edge-flux conservation

Varying the action gives the radial graph Poisson equation:

$$\left(\frac{1}{r} \right) \frac{d}{dr} \left[r \frac{d\Phi}{dr} \right] = -S(r)$$

Outside the visible source, $S(r)$ is approximately zero, so:

$$\frac{d}{dr} \left[r \frac{d\Phi}{dr} \right] = 0$$

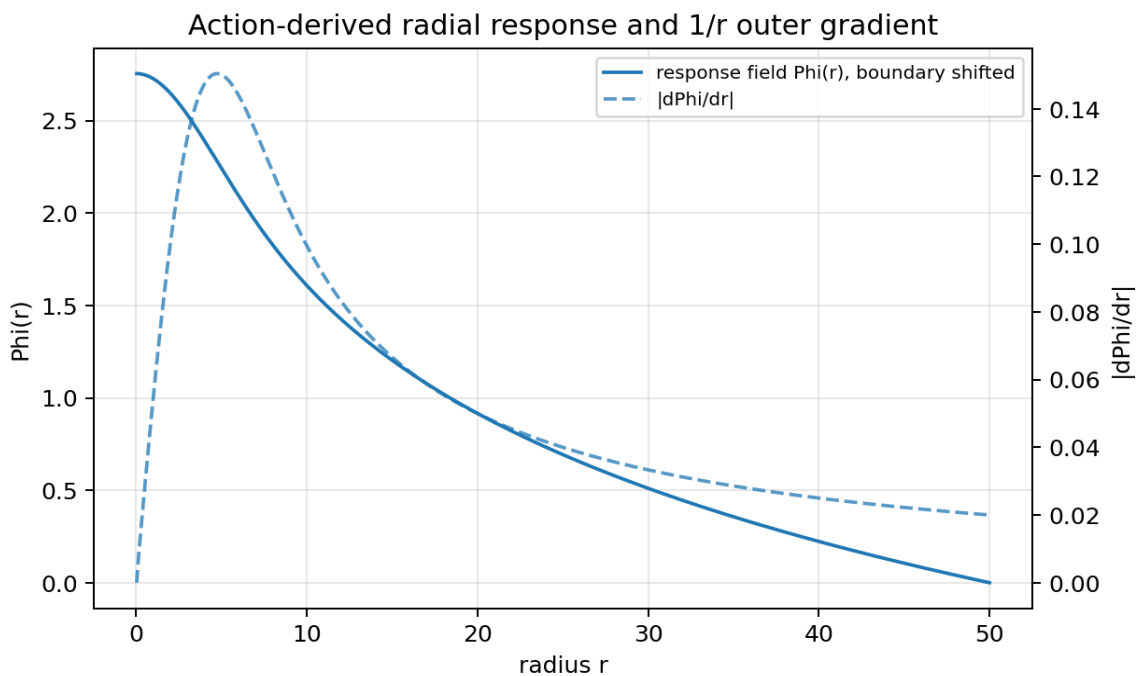
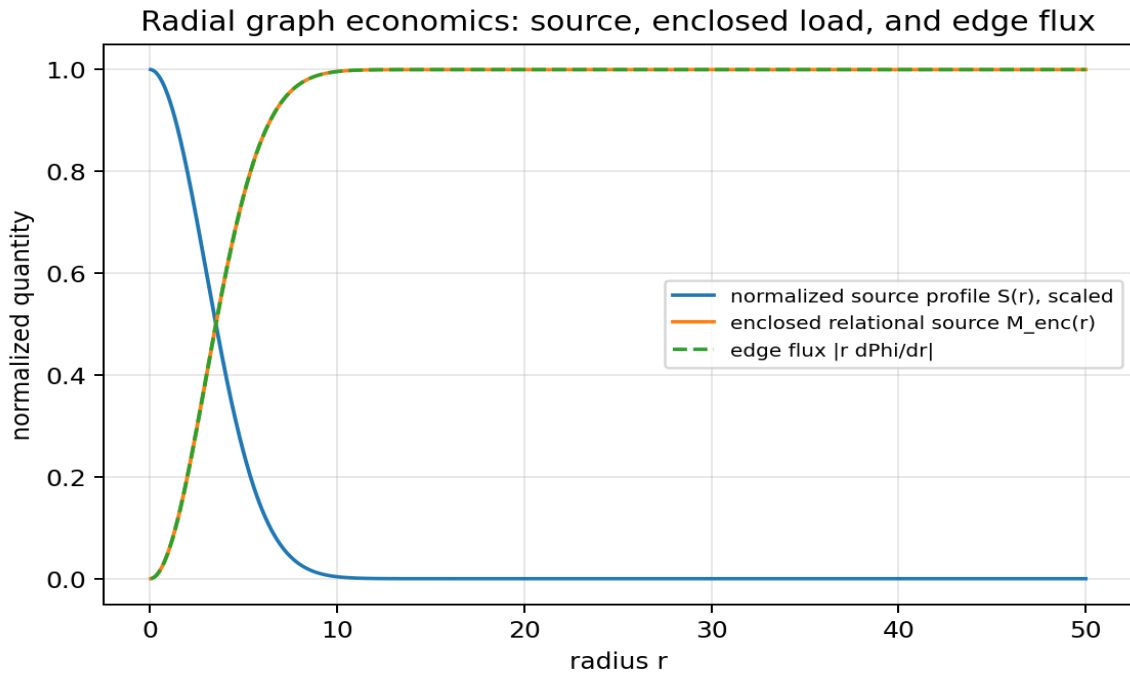
therefore:

$$r \frac{d\Phi}{dr} = \text{constant}$$

therefore:

$$\left| \frac{d\Phi}{dr} \right| \text{ proportional to } 1/r$$

This is the key 5B derivation. The flat-curve contribution is no longer inserted. It follows from edge-flux conservation on an effectively 2D annular graph.



4. Rotation-curve consequence

If relational acceleration is proportional to the edge-response gradient, then:

$$a_{rel}(r) \text{ proportional to } |d\Phi/dr| \text{ proportional to } 1/r$$

$$v_{rel}^2(r) = r a_{rel}(r)$$

therefore:

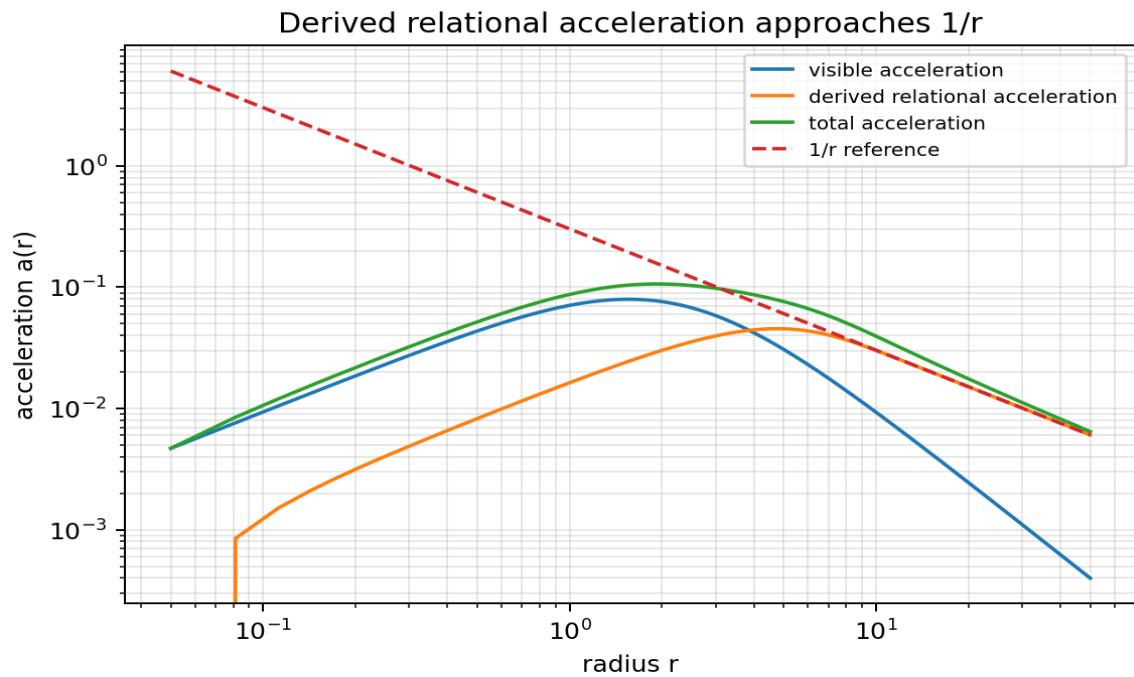
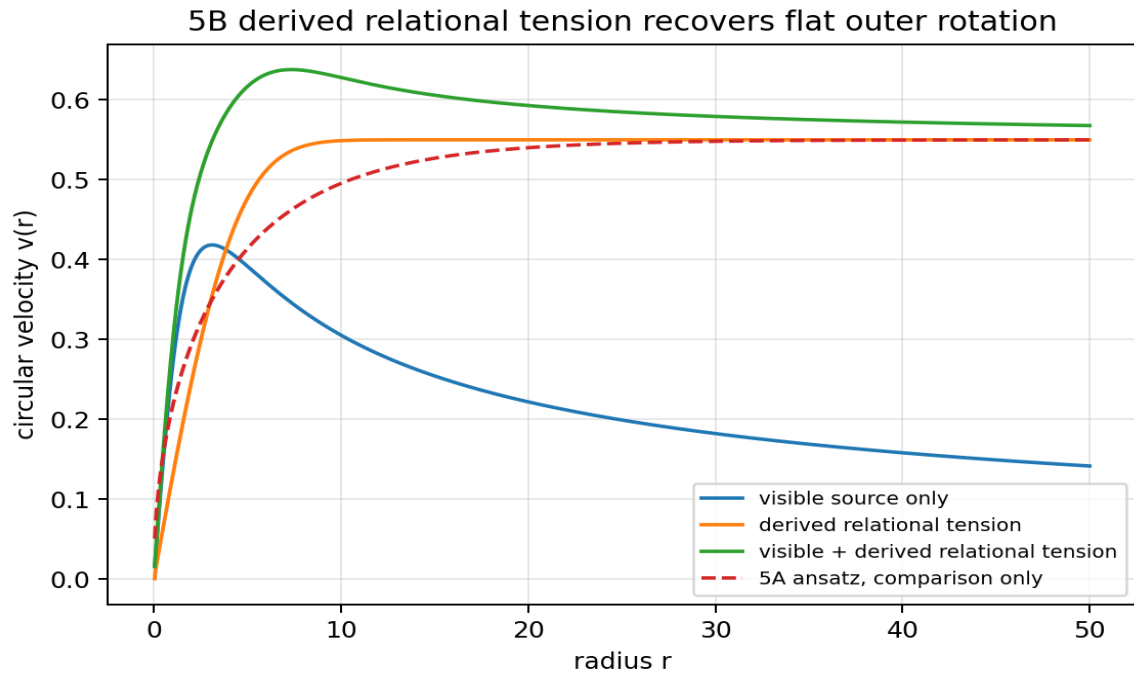
$$v_{rel}^2(r) \text{ approaches constant}$$

In the numerical model, the derived contribution is:

$$v_{rel}^2(r) = \Gamma M_{enc}(r)$$

$$M_{enc}(r) = \frac{\int_0^r S(s) s ds}{\int_0^{R_{max}} S(s) s ds}$$

The coupling Γ sets amplitude only. The radial shape comes from the graph action and the effective dimension of the annular network.



5. Numerical checks

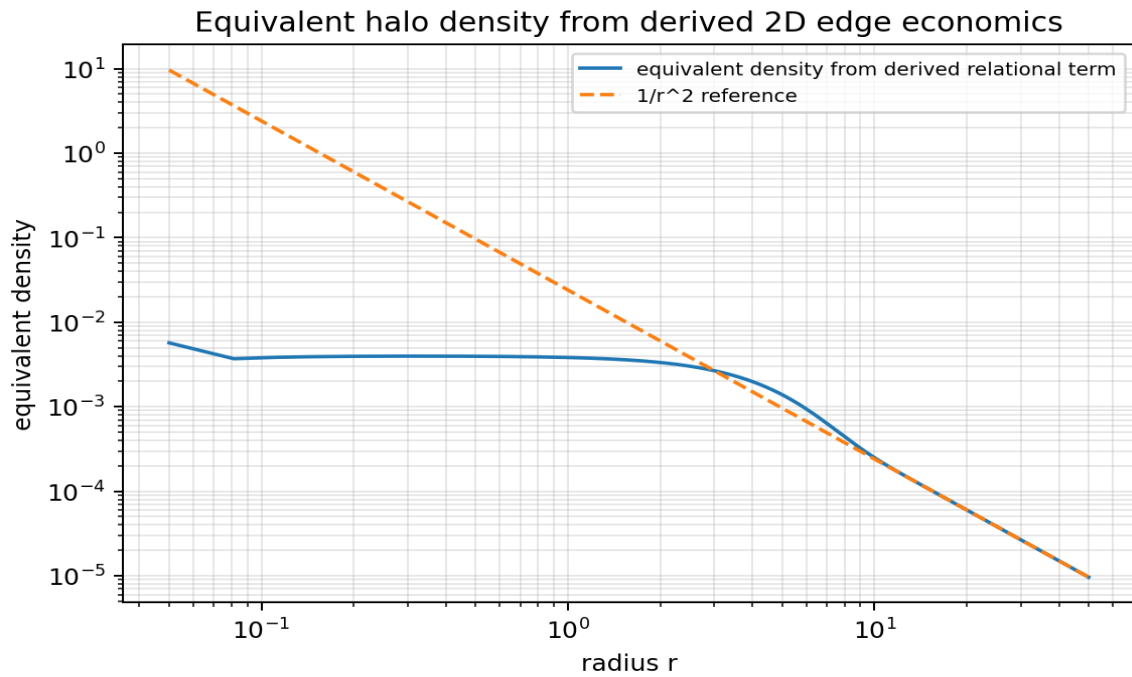
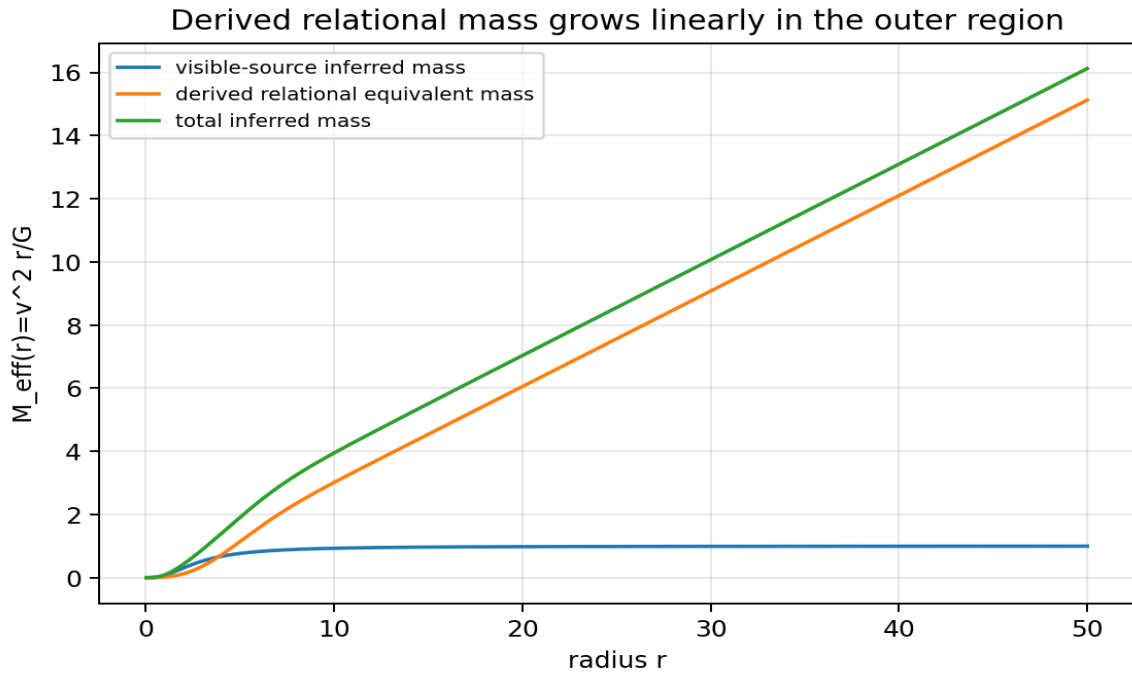
Diagnostic	Value	Interpretation
Euler residual max	8.886418e-06	Radial action equation solved accurately.
Euler residual RMS	1.706076e-06	Residual is small away from boundaries.
Outer edge-flux coefficient of variation	3.968530e-07	Outer $r d\Phi/dr $ is nearly constant.
Visible-only outer velocity slope	-0.4918	Kepler-like decline.
Derived relational velocity slope	0.0000	Flat contribution.
Total outer velocity slope	-0.0480	Nearly flat total curve.
Outer relational acceleration slope	-1.0000	Approaches $1/r$.
Relational mass outer slope	1.0000	Approaches M_{rel} proportional to r .
Relational mass linear R2	1.000000	Very close to linear outer growth.
Equivalent density outer slope	-2.0000	Near $1/r^2$ halo-like profile.

6. Effective missing-mass interpretation

A Newtonian observer who does not include relational tension would infer extra mass:

$$M_{\text{rel}}(r) = v_{\text{rel}}^2(r) \, r / G$$

Since v_{rel}^2 becomes constant, M_{rel} grows approximately linearly with r . That is the dark-matter-halo signature in this toy setting.



r	M_enc	v_b	v_rel	v_total	M_rel	rel. fraction
3.0	0.3934	0.4181	0.3450	0.5420	0.3570	0.405
5.0	0.7506	0.3916	0.4765	0.6168	1.1353	0.597
10.0	0.9961	0.3052	0.5489	0.6281	3.0133	0.764
15.0	1.0000	0.2541	0.5500	0.6059	4.5375	0.824
20.0	1.0000	0.2216	0.5500	0.5930	6.0500	0.860
30.0	1.0000	0.1818	0.5500	0.5793	9.0750	0.901
40.0	1.0000	0.1578	0.5500	0.5722	12.1000	0.924
50.0	1.0000	0.1412	0.5500	0.5678	15.1250	0.938

7. Why effective dimension matters

The derivation is not automatic in every graph dimension. In an effective radial graph dimension D , the source-free flux condition gives:

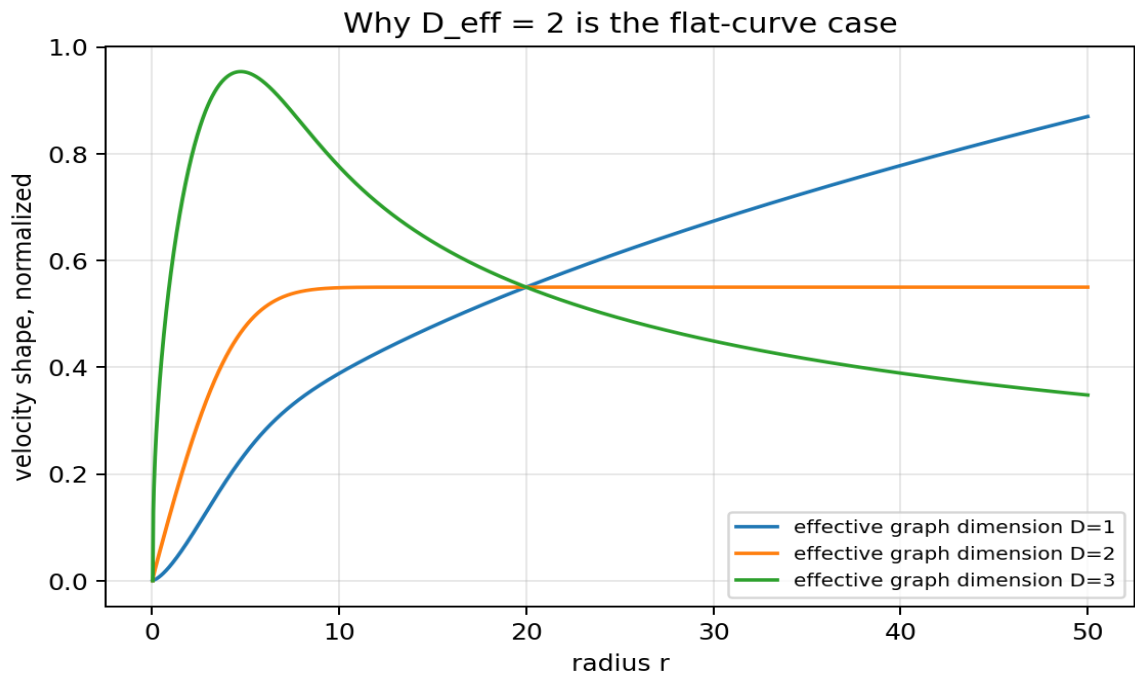
$$r^{(D-1)} \frac{d\Phi}{dr} = \text{constant}$$

therefore:

$$a(r) \propto \frac{1}{r^{(D-1)}}$$
$$v^2(r) = r a(r) \propto r^{(2-D)}$$

Therefore:

- $D=3$ gives v proportional to $r^{(-1/2)}$, the Keplerian decline.
- $D=2$ gives v approximately constant, the flat-rotation case.
- $D=1$ gives a rising curve.



Effective graph dimension D	Outer velocity slope	Behavior
1	0.5000	rising
2	0.0000	flat
3	-0.5000	Kepler-like decline

This is the most important guardrail of 5B: the flat curve emerges only if the relevant large-scale relational tension is effectively two-dimensional / annular / disk-like.

8. Interpretation inside V4.2

Toy Model 5B supports this limited ladder:

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radial annular graph action
-> edge-flux conservation
-> |dPhi/dr| proportional to 1/r
-> a_rel proportional to 1/r
-> v_rel^2 approximately constant
-> M_rel proportional to r
-> Newtonian observer infers halo-like missing mass
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It does not support this stronger ladder:

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graph edge economics
-> all dark matter solved
-> Lambda-CDM replaced
-> particle dark matter disproved
-> physical 3T proven
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9. What Toy Model 5B achieves

- It removes the explicit 5A flat-velocity ansatz.
- It derives the flat-curve shape from a native radial graph action.

- It shows edge-flux conservation gives the required $1/r$ acceleration in an effective 2D annular graph.
- It recovers M_{rel} proportional to r and a roughly $1/r^2$ equivalent density profile.
- It identifies effective graph dimension as the critical dark-sector condition.

10. What Toy Model 5B does not achieve

- It does not derive the coupling amplitude Γ or V_0 from first principles.
- It does not prove the galaxy-scale relational network is effectively 2D.
- It does not fit real galaxy rotation curves.
- It does not derive the baryonic Tully-Fisher relation.
- It does not address gravitational lensing, galaxy clusters, Bullet Cluster, CMB, or structure formation.
- It does not disprove particle dark matter.
- It does not prove physical 3T.

11. Close-out statement for 5B

Toy Model 5B is a stronger dark-sector analogue than 5A. The flat outer rotation contribution is not assumed directly; it emerges from minimizing a radial annular graph action. In an effective 2D relational graph, edge-flux conservation makes the relational gradient fall as $1/r$, which makes v_{rel}^2 approach a constant and makes the Newtonian-equivalent missing mass grow linearly with radius. The remaining open problem is not the shape; it is the origin of the amplitude and the physical reason why galactic-scale relational tension should be effectively two-dimensional.

12. Recommended next chapter

Recommended next: Toy Model 5C - Coupling Scale and Baryonic Tully-Fisher Check.

- Derive or constrain Γ from the baryonic source rather than choosing V_0 .
- Check whether v_{flat}^4 proportional to M_b can emerge.
- If not, identify what additional graph principle is required.
- Do not move to cluster lensing until the galaxy scaling law is tested.

13. Rebind prompt for continuation

We are resuming after Toy Model 5B.

5A:
Assumed a relational-tension velocity contribution:
 $v_{\text{rel}}^2 = v_{\text{inf}}^2 [1 - \exp(-r/R_t)]$
This flattened the rotation curve, but the term was inserted.

5B:
Removed the explicit flat-velocity ansatz.
Started from a radial annular graph action:

$$A[\Phi] = 2 \pi \int r \left[\frac{1}{2} \left(\frac{d\Phi}{dr} \right)^2 - S(r) \Phi(r) \right] dr$$

Variation:
 $(1/r) \frac{d}{dr} [r \frac{d\Phi}{dr}] = -S(r)$

Outside the source:
 $S \approx 0$
 $\frac{d}{dr} [r \frac{d\Phi}{dr}] = 0$
 $r \frac{d\Phi}{dr} = \text{constant}$
 $|d\Phi/dr| \propto 1/r$

If a_{rel} proportional to $|d\Phi/dr|$:
 a_{rel} proportional to $1/r$
 $v_{\text{rel}}^2 = r a_{\text{rel}} = \text{constant}$
 $M_{\text{rel}} = v_{\text{rel}}^2 r/G \propto r$

Numerical result:
Visible-only outer velocity slope approx -0.49.
Derived relational velocity slope approx 0.
Total outer velocity slope nearly flat.
Equivalent density approaches roughly $1/r^2$.
Outer M_{rel} grows linearly with radius.

Critical guardrail:
Flat rotation emerges only if the relevant relational graph is effectively 2D/annular.
The amplitude Γ / V_0 is still not derived.
No real galaxy data, lensing, CMB, clusters, Bullet Cluster, or structure formation tested.

Next recommended:
Toy Model 5C - Coupling Scale and Baryonic Tully-Fisher Check.